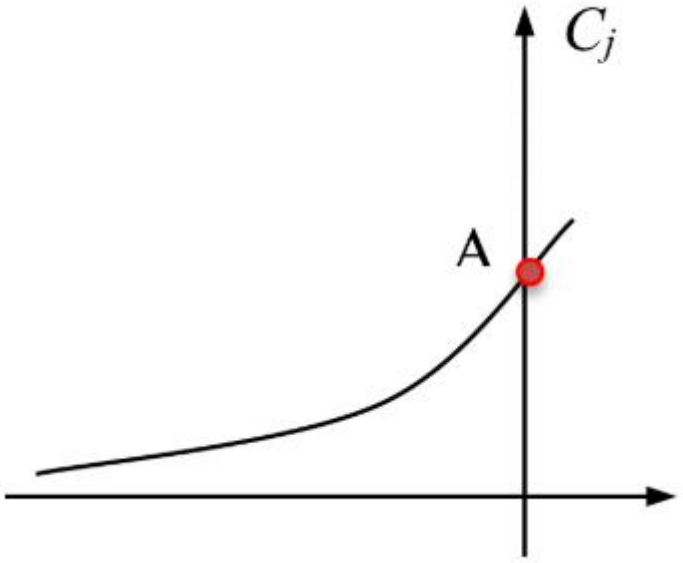
$W = \left\{ \frac{2\epsilon_0\epsilon_r(V_{bi}-V_{appl})}{e} \cdot \frac{N_a+N_d}{N_aN_d} \right\}^{\frac{1}{2}} = 3.64\times10^{-4}\,cm$

$= 3.64\,\mu m$

Homework 8 (due on 5/25/2026)

$V_{bi} = \frac{kT}{e} \ln \frac{N_aN_d}{n_i^2}$

1. In a p^+-n Si junction, the n side has a donor concentration of 10^{16} cm^{-3}. If $n_i = 10^{10}$ cm^{-3}, relative dielectric constant $\varepsilon_r = 12$, calculate the depletion width at a reverse bias of 100 V? Please plot the electrical field across the depletion region with Matlab? (Hint: Remember that p+ means very heavily doped, for example like 10^{19} cm^{-3}.) (20 points)
2. The following diagram shows the depletion capacitance of a p+/n junction. Assume the diode has an abrupt junction and p+ junction has a doping concentration about 1000 times higher than that of the n-side. (20 points)



$C_j = A \cdot \left\{ \frac{e\epsilon_s N_a N_d}{2(V_{bi}-V)(N_a+N_d)} \right\}^{\frac{1}{2}}$

$N_a = 10^3 N_d$

$\Rightarrow C_j \approx A \cdot \left\{ \frac{e\epsilon_s N_d}{2(V_{bi}-V)} \right\}^{\frac{1}{2}}$

① $N_a = 10^2 N_d'$

② $N_a' = 10^3 N_d$.

a) When the doping concentration of the n-side is increased by 10 times, what will be the value of the capacitance at zero bias (point A) be relative to the original one?

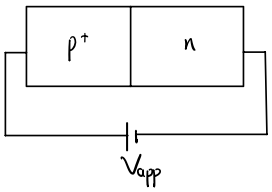
b) When the doping concentration of the p+ side is increased by 10 times, what will be the value of the capacitance at zero bias (point A) be relative to the original one?

3. A semiconductor with a band gap of 0.8 eV and an intrinsic carrier concentration of 10^{12} cm^{-3} is doped with 10^{18} cm^{-3} donors on the left half and 10^{17} cm^{-3} acceptors on the right half. Draw the equilibrium band diagram. Plot the band diagram with Matlab, calculate the junction built-in potential and the position of the Fermi level, and indicate them on the band diagram. (20 points)



4. For the structure like p-n1-n2, where p=10^{18} cm^{-3}, n1=10^{15} cm^{-3}, n2=5x10^{17} cm^{-3}, the thickness of p layer is 300nm, the thickness of n1 layer is 200nm, the thickness of n2 layer is 500nm. (40 points)
   a) Plot the band diagram under depletion approximation at 0V. Does n1 layer fully depleted?
   b) Does n1 layer fully depleted if n1 = 5x10^{16} cm^{-3}? If not, how much additional voltage should be added to fully deplete the n1 layer?

1.



$N_A = 10^{19}\ cm^{-3}$. $N_D = 10^{16}\ cm^{-3}$. $V_{app} = -100\ V$. $\varepsilon_0 = 8.85 \times 10^{-14}\ F/cm$

Consider $T = 300\ k$

$$V_{bi} = \frac{kT}{e} \cdot \ln\frac{N_A N_D}{n_i^2} = 0.894\ V$$

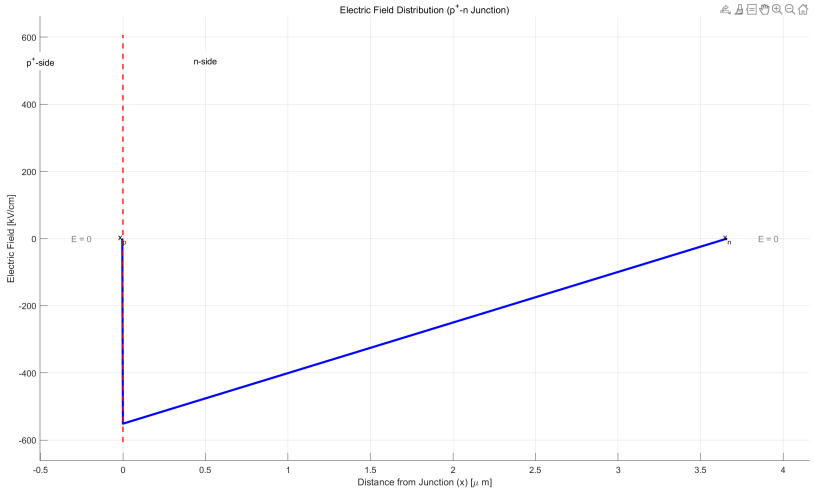
$$x_p = \sqrt{\frac{2\varepsilon_0\varepsilon_r (V_{bi} - V_{app})}{e}} \sqrt{\frac{N_D}{N_A (N_A + N_D)}} = 3.66 \times 10^{-7}\ cm$$

$$x_n = \sqrt{\frac{2\varepsilon_0\varepsilon_r (V_{bi} - V_{app})}{e}} \sqrt{\frac{N_A}{N_D (N_A + N_D)}} = 3.66 \times 10^{-4}\ cm$$

Depletion region width: $W = x_n + x_p = 3.66 \times 10^{-4}\ cm = 3.66\ \mu m$

$$\rho(x) = \begin{cases} -eN_A, & -x_p \le x < 0 \\ eN_D, & 0 < x \le x_n \end{cases}$$

$$\frac{\partial E}{\partial x} = \frac{\rho(x)}{\varepsilon_0\varepsilon_r} = \begin{cases} -1.51 \times 10^{12}\ V/cm^2, & -x_p \le x < 0 \\ 1.51 \times 10^{9}\ V/cm^2, & 0 < x \le x_n \end{cases}$$



2. Assume $N_A = 1000\,N_0$. $N_D = N_0$

$$C' = \left\{ \frac{e\epsilon_s N_A N_D}{2(V_{bi} - V_A)(N_A + N_D)} \right\}^{\frac{1}{2}}$$

At $V_A = 0$, $C' = \left\{ \frac{e\cdot\epsilon_s \cdot 1000 N_0}{2V_{bi}\cdot 1001} \right\}^{\frac{1}{2}}$

(a) $N_A = 1000\,N_0$ $N_D = 10\,N_0$. $C_1' = \left\{ \frac{e\epsilon_s \cdot 1000 N_0}{2V_{bi}\cdot 101} \right\}^{\frac{1}{2}}$

$$\frac{C_1'}{C'} = \sqrt{\frac{1001}{101}} \approx 3.15$$

The capacitance at the origion is 3.15 times the origional ones.

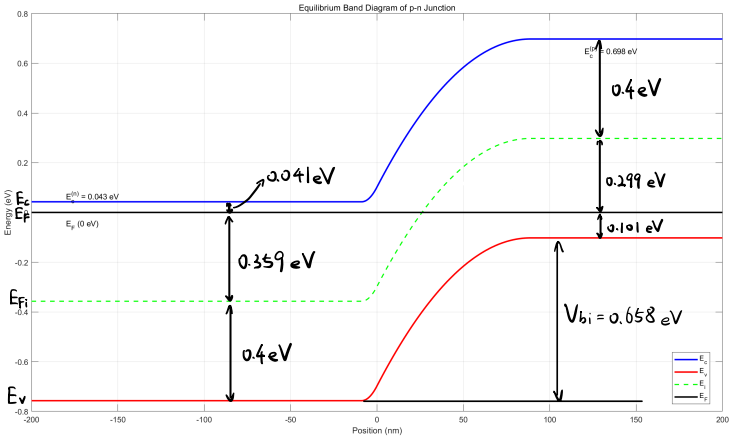
(b) $N_A = 10000 N_0$. $N_D = N_0$. $C_2' = \left\{ \frac{e\epsilon_s \cdot 10000 N_0}{2V_{bi}\cdot 10001} \right\}^{\frac{1}{2}}$

$$\frac{C_2'}{C'} = \sqrt{\frac{10000}{10001}} \approx 1$$

The value of capacitance at the origion is almost the same with the original ones.

3. $E_g = 0.8$ eV, $n_i = 10^{12}\,cm^{-3}$

$N_D = 10^{18}\,cm^{-3}$	$N_A = 10^{17}\,cm^{-3}$



left: $E_F - E_{Fi} = |\varphi_n| = \frac{kT}{e}\cdot \ln\frac{N_D}{n_i}$

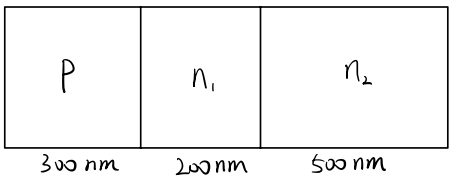
$$E_F - E_{Fi} = |\varphi_n| = 0.359 \text{ eV}$$

right: $E_{Fi} - E_F = |\varphi_p| = \frac{kT}{e}\cdot \ln\frac{N_A}{n_i}$

$$E_{Fi} - E_F = |\varphi_p| = 0.299 \text{ eV}$$

$$V_{bi} = |\varphi_n| + |\varphi_p| = 0.658 \text{ eV}$$

4.



$P = 10^{18}\,cm^{-3}$, $n_1 = 10^{15}\,cm^{-3}$, $n_2 = 5\times10^{17}\,cm^{-3}$, Assume $T = 300k$, $n_i = 10^{10}\,cm^{-3}$, $\varepsilon_r = 12$

a) Region P: $N_A = 10^{18}\,cm^{-3}$. $E_{Fi} - E_{Fo} = kT\cdot\ln\frac{N_A}{n_i}$

$\Rightarrow E_{Fi} - E_{F_1} = 0.1796\ eV$

Region n_1: $N_{D1} = 10^{15}\,cm^{-3}$ $E_{F_1} - E_{Fi} = kT\cdot\ln\frac{N_{D1}}{n_i}$

$\Rightarrow E_{F_1} - E_{Fi} = 0.2993\ eV$

Region n_2: $N_{D_2} = 5\times10^{17}\,cm^{-3}$ $E_{F_2} - E_{Fi} = kT\cdot\ln\frac{N_{D_2}}{n_i}$

$\Rightarrow E_{F_2} - E_{Fi} = 0.4609 eV$

Region P and n_1: $V_{bi1} = \frac{kT}{e}\cdot\ln\frac{N_{D_1}\cdot N_{D_2}}{n_i^2} = 0.7783\ V$

$$x_p = \sqrt{\frac{2\varepsilon_0\varepsilon_r V_{bi1}}{e}\cdot\frac{N_{D1}}{N_A(N_A+N_{D1})}} = 1.015\ nm$$

$$x_{n_1} = \sqrt{\frac{2\varepsilon_0\varepsilon_r V_{bi1}}{e}\cdot\frac{N_A}{N_{D1}(N_A+N_{D1})}} = 1015\ nm$$

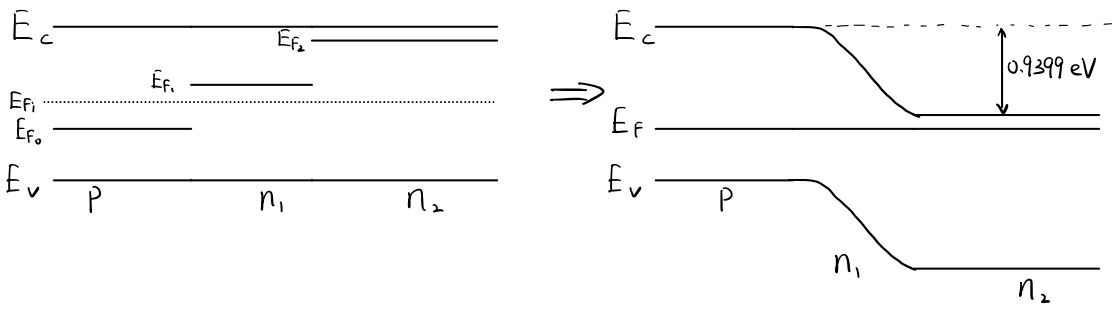
Region n_1 and n_2: $V_{bi2} = \frac{E_{F_2} - E_{F_1}}{e} = \frac{kT}{e}\cdot\ln\frac{N_{D_2}}{N_{D_1}} = 0.1616\ V$

Depletion approximation: $\rho(x) = \begin{cases} \frac{-e\cdot N_{D_1}}{\varepsilon_0\varepsilon_r} & \text{Region } n_1 \\ \frac{eN_{D_2}}{\varepsilon_0\varepsilon_r} & \text{Region } n_2 \end{cases}$

$$\Rightarrow x_{n_1}' = \sqrt{\frac{2\varepsilon_0\varepsilon_r V_{b2}}{e}\cdot\frac{N_{D1}}{N_{D_2}(N_{D_1}+N_{D_2})}} = 0.9257\ nm$$

$$x_{n_2}' = \sqrt{\frac{2\varepsilon_0\varepsilon_r V_{b2}}{e}\cdot\frac{N_{D2}}{N_{D_1}(N_{D_1}+N_{D_2})}} = 462.9\ nm$$

$W_{n_1} = x_{n_1} + x_{n_1}' = 1015\ nm > 200\ nm$, n_1 fully depleted.



b) $n_1 = 5 \times 10^{16}\ cm^{-3}$

Region P and n_1 : $V_{bi1} = \frac{kT}{e} \cdot \ln \frac{N_{D_1} N_{D_2}}{n_i^2} = 0.8620\ V$

$x_{n_1} = \sqrt{\frac{2\varepsilon_0 \varepsilon_r V_{bi1}}{e} \cdot \frac{N_A}{N_{D_1}(N_A + N_{D_1})}} = 151.2\ nm$

Region n_1 and n_2 : $V_{bi2} = \frac{E_{F_2} - E_{F_1}}{e} = \frac{kT}{e} \cdot \ln \frac{N_{D_2}}{N_{D_1}} = 0.05987\ V$

$x_{n_1}' = \sqrt{\frac{2\varepsilon_0 \varepsilon_r V_{b2}}{e} \cdot \frac{N_{D_1}}{N_{D_2}(N_{D_1} + N_{D_2})}} = 3.799\ nm$

$W_{n_1} = x_n + x_{n_1}' = 155\ nm < 200\ nm$. Not fully depleted.

$W_{n_1} = \sqrt{\frac{2\varepsilon_0 \varepsilon_r (V_{bi1} - V_{app})}{e} \cdot \frac{N_A}{N_{D_1}(N_A + N_{D_1})}} + \sqrt{\frac{2\varepsilon_0 \varepsilon_r (V_{b2} - V_{app})}{e} \cdot \frac{N_{D_1}}{N_{D_2}(N_{D_1} + N_{D_2})}} = 200\ nm$

$\Rightarrow V_{app} \approx -0.67\ V$

We should add 0.67 additional reverse bias.

